

(1001CJA621620049)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

11-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	A	C	B	D	B,C	A,D	C,D	A,C
	Q.	11	12								
	A.	B,D	A,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	10.25 to 10.36	3.70 to 3.80	90.00	67.00 to 69.00	10.00	55.00				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	B	D	C	B	B,C	A,B	A,B,C,D	A,C,D
	Q.	11	12								
	A.	A,C	C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	74.00	2.00	3.00	0.92 to 0.96	4.00	0.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	C	D	C	C	A,B,C,D	A,D	B,D	B,C
	Q.	11	12								
	A.	A,D	B,C,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	0.00	7.00	1.12	0.20	36.00	194.00				

(1001CJA621620050)

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(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

11-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

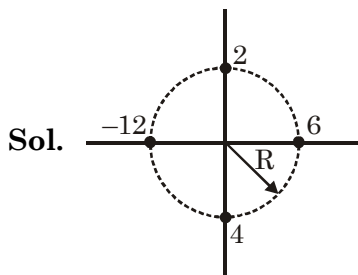
SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,B	A,B,D	A,B,C,D	A,C	A,B	A,B,C				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	6768.00	4500.00	1.07	2.44 to 2.45	0.49 to 0.50	2.63 to 2.64				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	0	1	5	1	9	5				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6				
	A.	C,D	A,B,C,D	A,C	A,B,C,D	A,C,D	A,B				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	53.00	5.00	4.00	2.00	3.00	2.15				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	0	4	5	3	3	3				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,D	B,D	A,D	A,B,C,D	A,C	B,C,D				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	60.00	136.00	5049.00	1.50	482.00	5150.00				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	4	1	2	7	9	8				

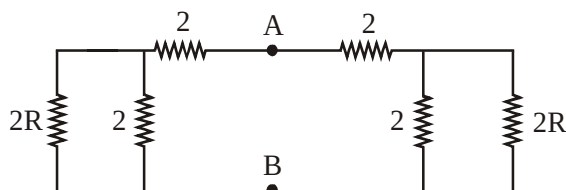
PAPER-1**PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (C)**

$$V_{\text{at } (0, 0, z)} = 0$$

2. **Ans. (A)**

Sol. The equivalent circuit can be redrawn as shown.

$$\text{Let } R = \frac{2R \times 2}{2R + 2} + 2$$



$$R = \frac{2R}{R+1} + 2$$

$$R = \frac{2R + 2R + 2}{(R+1)}$$

$$\Rightarrow R = \frac{2 + 4R}{R+1}$$

$$R^2 - 3R - 2 = 0$$

$$R = \frac{3 + \sqrt{17}}{2}$$

$$R_{AB} = \frac{R}{2} = \frac{(3 + \sqrt{17})}{4}$$

3. **Ans. (A)**

$$\text{Sol. } (R + r)\omega = r\omega'$$

$$(R + r)\omega dt = r\omega' dt$$

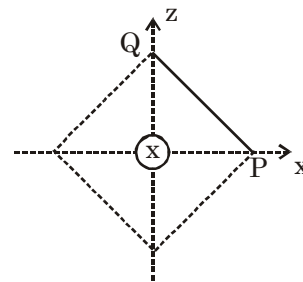
$$\frac{R + r}{r} = \frac{\omega' dt}{\omega dt}$$

4. **Ans. (C)**

$$\text{Sol. For wire along z axis } \int \vec{B} \cdot d\vec{\ell} = 0 \quad (B \perp d\ell)$$

$$\text{For wire along y axis } \oint \vec{B} \cdot d\vec{\ell} = \mu_0 i$$

$$\int_Q^P \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 i}{4}$$

5. **Ans. (B)**

$$\text{Sol. } \frac{\lambda}{2} = 6.77 \text{ cm}$$

$$v = f\lambda$$

$$\sqrt{\frac{rRT}{M}} = f\lambda$$

$$r = 1.4$$

6. **Ans. (D)**

$$\text{Sol. } \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mk}}$$

$$k = \frac{h^2}{2m\lambda^2}$$

$$\Delta k = \frac{h^2}{2m} \left(\frac{1}{\lambda_1^2} - \frac{1}{\lambda_2^2} \right)$$

$$\Delta k = 5.6 \text{ eV}$$

7. **Ans. (B,C)**

Sol.

$$V_L = V_C = V_R$$

$$\Rightarrow x_L = x_C = R$$

when inductor is short circuited

$$Z = \sqrt{R^2 + x_C^2} = \sqrt{2} R$$

$$\therefore i = \frac{30}{Z} = \frac{30}{\sqrt{2} R}$$

$$\therefore V_L = ix_L = \frac{30}{\sqrt{2} R} \times R = \frac{30}{\sqrt{2}}$$

$\therefore$ (A) is incorrect and with similar calculations (B) will be correct.

Here f_0 is the resonance frequency as $v_L = v_C$

$$\Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$\text{and } \omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

$$\therefore \frac{x_L}{x_C} = \frac{\omega L}{1/\omega C} = \omega^2 LC$$

$$\text{Given } f = 2f_0$$

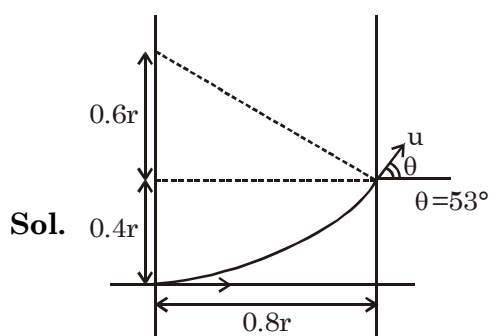
$$\Rightarrow \omega = 2\omega_0$$

$$\therefore \frac{x_L}{x_C} = 4$$

$\therefore$ (C) is also correct

$\therefore$ (B), (C)

8. **Ans. (A,D)**



$$y - 0.4r = \frac{4}{3}(x - 0.8r)$$

$$y = 1.6r + 0.4r$$

$$y = 2r \text{ (Passes through C)}$$

9. **Ans. (C,D)**

$$\text{Sol. } -4 - L \frac{di}{dt} + \frac{2}{4} = 0$$

$$L \frac{di}{dt} = -35 \text{ volt} \Rightarrow i \text{ (decreasing)}$$

10. **Ans. (A,C)**

$$\text{Sol. } RB \rightarrow 1\Omega$$

$$i = \frac{10}{13}$$

$$\Delta v = \frac{10}{13} \times 10 = \frac{100}{13}$$

$$7 = \frac{100}{13} \times \frac{x}{4} \Rightarrow x = \frac{13 \times 7}{25} = 3.64$$

for accuracy obtain null point at maximum distance (take $RB = 2$)

$$i = \frac{10}{14}$$

$$\Delta v = \frac{10}{14} \times 10 = \frac{100}{14} \Rightarrow 7 = \frac{100}{14} \times \frac{x}{4}; x = 3.92$$

11. **Ans. (B,D)**

Sol. When water does not overflow, normal is increased due to buoyant force when water overflow, weight of liquid overflow = Buoyant force so normal remains same.

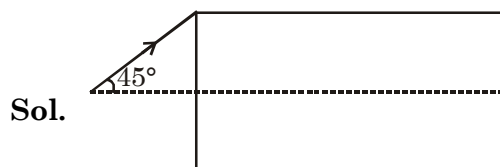
12. **Ans. (A,D)**

$$\text{Sol. } \sigma AT_s^4 = 72$$

$$\lambda T = \text{constant} = 2.88 \times 10^{-3}$$

SECTION-II

1. **Ans. 10.25 to 10.36**



$$\text{Here, } A > 2\theta_c$$

$$\frac{\Omega}{4\pi} = \frac{2\pi(1 - \cos\theta)}{4\pi} = \frac{1 - \cos\theta}{2} = \frac{1 - \frac{1}{\sqrt{2}}}{2}$$

$$P = \frac{\sqrt{2}-1}{2\sqrt{2}} \times 50\sqrt{2} = 25(\sqrt{2}-1)$$

2. Ans. 3.70 to 3.80

$$\text{Sol. } \eta = \frac{P_0 V_0 - \frac{\pi P_0 V_0}{4}}{2P_0 V_0 - \frac{\pi P_0 V_0}{4} + \frac{3}{2} + 3P_0 V_0}$$

$$= \frac{4-\pi}{26-\pi}$$

$$= 3.76$$

3. Ans. 90.00

$$\text{Sol. } mg = m(\theta_0 \omega)^2 d$$

$$\omega = \frac{1}{\theta_0} \sqrt{\frac{g}{d}}$$

4. Ans. 67.00 to 69.00

$$\text{Sol. } 18000 = \frac{dN}{A dt} = \frac{\lambda N_0}{A}$$

$$1.8 \times 10^4 \times 4\pi R^2 = \frac{0.693}{6930 \times \pi \times 10^{11}}$$

$$n = \frac{4\pi \times 10^{15} \times 1.8 \times 6.4^2 \times 10^6}{6 \times 10^{23}}$$

$$= 4 \times 3 \times 6.4^2 \times 10^4 \times \frac{14 \times 10^{-3}}{100} \times 10$$

$$= 68.81$$

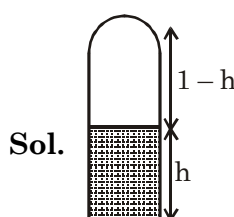
5. Ans. 10.00

$$\text{Sol. } T = 0$$

(Free fall)

$$a = g$$

6. Ans. 55.00



Sol.

$$P_1 V_1 = P_2 V_2$$

$$10^5(0.25) = (10^5 - 10^2 \times 10 \times h)(1-h)$$

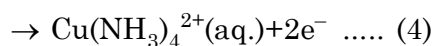
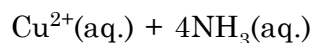
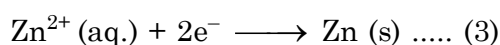
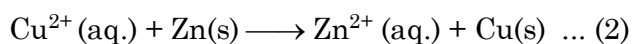
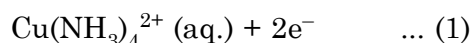
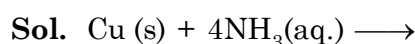
$$h \approx 0.75 \Rightarrow \text{no liquid falls}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(C)



$$(1) = (4) - (2) - (3)$$

$$-2FE^0 = -RT \ln 10^{20} + 2F(1.1) - 2F(.76)$$

$$E^0 = \frac{2.303RT}{2F} \log 10^{20} - 0.34$$

$$\Rightarrow E^0 = \frac{0.06}{2} \times 20 - 0.34$$

$$= 0.6 - 0.34 = 0.26 \text{ V}$$

2. Ans. (D)

3. Ans.(B)

4. Ans.(D)

Sol. (A) For solution Z_2 at P_1 pressure

$$X_A = 0.25 \quad X_B = 0.75$$

$$Y_A = 0.5 \quad Y_B = 0.5$$

(B) For solution Z_2 at P_3 pressure $\rightarrow$ solution will not vapourise so

$$X_A = 0.4 ; X_B = 0.6$$

(C) For solution Z_1 at P_2 pressure $\rightarrow$ solution will not vapourise so

$$X_A = 0.2 ; X_B = 0.8$$

5. Ans. (C)

6. Ans.(B)

7. Ans.(B, C)

Sol. For 'Q' using steady state concept

Rate of appearance of Q = Rate of disappearance of Q

$$K_1[P] = K_2[Q]$$

Number of nuclei of Q

$$= \frac{k_1}{k_2} \times \text{no. of nuclei of P}$$

$$= \frac{10/60}{1000} \times 6 \times 10^{23} = 10^{20}$$

For 'R'

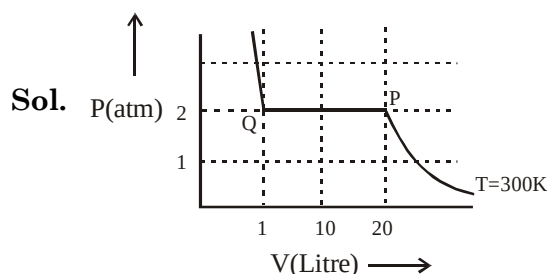
Since rate constant is very high for second step than first step

Number of nuclei of R = Number of nuclei of P disintegrated = 6×10^{23}

8. Ans. (A, B)

9. Ans.(A,B,C,D)

10. Ans.(ACD)



(A) A real gas condense if external pressure is just greater than vapour pressure (B,C,D)

At point 'P' whole real gas is in gaseous phase so density of gas

$$= \frac{1000}{20 \times 1000} = 0.05 \text{ gm/ml}$$

At point 'Q' whole substance is in liquid phase so density of liquid

$$= \frac{1000}{1 \times 1000} = 1 \text{ gm/ml}$$

Note : during condensation at 300K density of gas and liquid remains constant, only amount varies.

11. Ans. (A, C)

12. Ans.(C)

SECTION-II

1. Ans. (74.00)

State-1	State-2	State-3
0.5 atm	1.0 atm	1 atm
80 litre	80 litre	20 litre
500 K	1000 K	250 K
1 mole	1 mole	1 mole

$$(q_{\text{sys}})_{1\text{-step}} = nC_V(\Delta T)$$

$$= 1 \times \frac{5}{2} R (1000 - 500) = \frac{2500}{2} R$$

$$(\Delta S_{\text{surro.}})_{\text{step-1}} = -\left(\frac{q_{\text{sys.}}}{T}\right) = \frac{-\frac{2500}{2} R}{1000} = \frac{-2.5}{2} R$$

$$(q_{\text{sys}})_{2\text{-step}} = nC_p(\Delta T)$$

$$= 1 \times \frac{7}{2} R (250 - 1000) = \frac{-5250}{2} R$$

$$(\Delta S_{\text{surro.}})_{\text{step-2}} = -\left(\frac{q_{\text{sys.}}}{T}\right) = \frac{\frac{5250}{2} R}{250}$$

$$= \frac{21}{2} R$$

$$(\Delta S_{\text{surro.}}) = \frac{-2.5}{2} R + \frac{21}{2} R = \frac{18.5}{2} R$$

$$= \frac{18.5}{2} \times 0.08 \times 100 = 74 \text{ J/K}$$

2. Ans. (2.00)

3. Ans.(3.00)

4. Ans.(0.92 to 0.96)

Sol. $PF = \frac{1 \times \pi r^2 + 1 \times \pi[(\sqrt{2}-1)r]^2}{(2r)^2} = 0.96$

5. Ans. (4.00)

6. Ans.(0.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A)**

Let D is origin and position vectors of A, B, C are $\vec{a}, \vec{b}, \vec{c}$ respectively, then $|\vec{a}| = |\vec{b}| = |\vec{c}| = \ell$ and

$$|\vec{a} \cdot \vec{b}| = |\vec{b} \cdot \vec{c}| = |\vec{c} \cdot \vec{a}| = \ell^2 \cos 60^\circ = \frac{\ell^2}{2}$$

$$\text{Now } [\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \frac{\ell^6}{2}$$

now volume of tetrahedron = $\frac{1}{3} \times \text{Area of base} \times \text{height}$

$$\Rightarrow \frac{1}{6} [\vec{a} \vec{b} \vec{c}] = \frac{1}{3} \times \left(\frac{\sqrt{3}}{4} \ell^2 \right) \times \text{height}$$

$$\Rightarrow \text{height} = \frac{\frac{\ell^3}{\sqrt{3}} \times \frac{1}{2}}{\frac{\sqrt{3}}{4} \ell^2} = \frac{\sqrt{2}}{\sqrt{3}} \ell$$

2. **Ans. (D)**

Sol. Apply LMVT

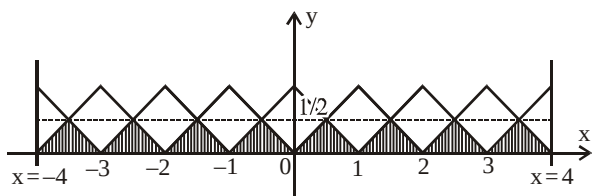
$$f'(x) = \frac{f(5) - f(-2)}{5 - (-2)} \text{ for some } x \text{ in } (-2, 5)$$

$$\text{Now, } -4 \leq \frac{f(5) - f(-2)}{7} \leq 3$$

$$-28 \leq f(5) - f(-2) \leq 21$$

$\therefore$ Greatest possible value of $f(5) - f(-2)$ is 21.

3. **Ans. (C)**

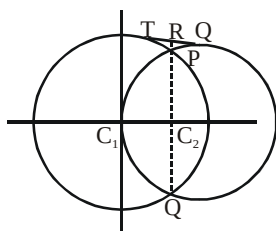


$$\text{Shaded area} = \text{required area} = 8 \times \frac{1}{2} \times \frac{1}{2} = 2 \text{ sq. units}$$

4. **Ans. (D)**

$$P\left(\frac{9}{4}, \frac{\sqrt{63}}{4}\right)$$

$$Q\left(\frac{9}{4}, -\frac{\sqrt{63}}{4}\right)$$



$$QT = \sqrt{(C_1 C_2)^2 - (r_1 - r_2)^2} = \sqrt{3}$$

$$(RT)^2 = RP \times RQ$$

$$\text{Let } RP = x \Rightarrow \frac{3}{4} = x \left(x + \frac{\sqrt{63}}{2} \right)$$

$$\Rightarrow x = \frac{-\sqrt{63} + \sqrt{75}}{4}$$

$$RP + RQ = \left(\frac{-\sqrt{63} + \sqrt{75}}{2} \right) + \frac{\sqrt{63}}{2}$$

$$= \frac{\sqrt{75}}{2} = \sqrt{\frac{75}{4}}$$

5. **Ans. (C)**

$$v = \begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 2 \\ 1 & 3 & 1 \end{vmatrix} [\vec{a} \vec{b} \vec{c}]$$

$$= (-14 + 1 + 4) \begin{vmatrix} 1 & 3 & 5 \\ 3 & 5 & 7 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= (-9)[-12 + 30 - 10] = -72$$

6. **Ans. (C)**

Equation of the tangent at the vertex is

$$x - y + 1 = 0 \quad \dots(i)$$

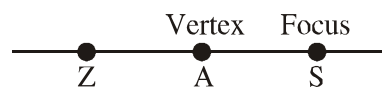
Equation of the axis of the parabola is

$$x + y + k = 0 \quad \dots(ii)$$

Since it passes through (0, 0) $0 + 0 + k = 0$

$$\Rightarrow k = 0$$

$$\therefore \text{Equation of axis is } x + y = 0 \quad \dots(iii)$$



Where z is a point on directrix

$$\text{Solving (i) and (iii) we get } A\left(\frac{-1}{2}, \frac{1}{2}\right)$$

$$\therefore z \text{ is } (-1, 1)$$

$$\text{Now directrix is } x - y + c = 0 \quad \dots(iv)$$

But equation (iv) is passes through z

$$-1 - 1 + c = 0$$

$$\Rightarrow c = 2$$

$$\text{So directrix is } x - y + 2 = 0$$

$$\text{Using } PS = PM$$

$$PS^2 = PM^2$$

7. Ans. (A,B,C,D)

$$\phi'(x) = 3f'\left(\frac{x^2}{3}\right) \cdot \frac{2x}{3} - f'(3-x^2)(2x)$$

$$= 2x \left(f'\left(\frac{x^2}{3}\right) - f'(3-x^2) \right)$$

For $\phi(x)$ to be increasing

$$(I) \text{ If } x > 0 \text{ then } f'\left(\frac{x^2}{3}\right) > f'(3-x^2)$$

$$\Rightarrow \frac{x^2}{3} > 3-x^2 \quad (f''(x) > 0)$$

$$x \in \left(-\infty, -\frac{3}{2}\right) \cup \left(\frac{3}{2}, \infty\right) \Rightarrow x \in \left(\frac{3}{2}, 4\right)$$

Similarly for decreasing $\phi(x)$

$$x > 0, x \in \left(-\frac{3}{2}, \frac{3}{2}\right) \Rightarrow x \in \left(0, \frac{3}{2}\right)$$

$$(II) \text{ If } x < 0 \text{ } f'\left(\frac{x^2}{3}\right) < f'(3-x^2)$$

$$\Rightarrow \frac{x^2}{3} < 3-x^2 \Rightarrow x \in \left(-\frac{3}{2}, \frac{3}{2}\right)$$

$$\Rightarrow x \in \left(-\frac{3}{2}, 0\right)$$

$$\text{If } x < 0, x \in \left(-\infty, -\frac{3}{2}\right) \cup \left[\frac{3}{2}, \infty\right)$$

$$\Rightarrow x \in \left(-3, -\frac{3}{2}\right)$$

8. Ans. (A,D)

Let A_i : Number i is selected from A.

B_j : Number j is selected from B.

$$\Rightarrow p = P(n(A) \leq n(B)) = 1 - P(n(A) > n(B))$$

$$= 1 - \{P(A_2).P(B_1) + P(A_3).P(B_1 \text{ or } B_2)\}$$

$$= 1 - \left\{ \frac{1}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{2}{5} \right\} = \frac{4}{5} = P(n(B) \leq n(A))$$

$$= P(A_1).P(B_1) + P(A_2).P(B_1 \text{ or } B_2) + P(A_3).P(B_1 \text{ or } B_2 \text{ or } B_3)$$

$$= \frac{1}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{2}{5} + \frac{1}{3} \times \frac{3}{5}$$

$$q = \frac{2}{5}$$

9. Ans. (B,D)

$$P(x) = (x-1)^2 Q_1(x) + 1$$

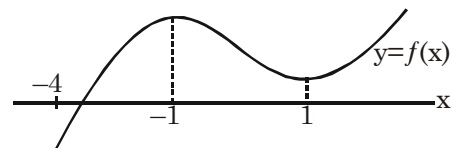
$$\text{and } P(x) = (x+1)^2 Q_2(x) + 3$$

$$P(1) = 1, P(-1) = 3, P'(1) = 0 = P'(-1)$$

$$\therefore P'(x) = \lambda(x^2 - 1)$$

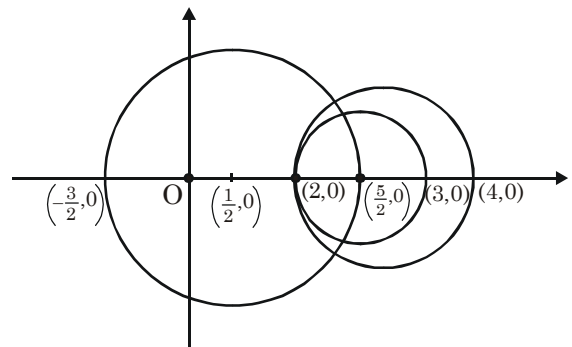
$$\Rightarrow P(x) = \lambda \left(\frac{x^3}{3} - x \right) + \mu$$

$$P(x) = \frac{x^3}{2} - \frac{3}{2}x + 2$$



10. Ans. (B,C)

$$\text{Sol. } \left| z_1 - \frac{1}{2} \right| = 2; \left| z_2 - \frac{5}{2} \right| = \frac{1}{2}; |z_3 - 3| = 1$$



from figure,

$$|z_1 + z_2 + z_3|_{\max} = 4 + 3 + \frac{5}{2} = \frac{19}{2}$$

$$|z_1 + z_2 + z_3|_{\min} = 2 + 2 - \frac{3}{2} = \frac{5}{2}$$

11. Ans. (A,D)

$$\text{Sol. } y \cdot (y^1)^2 + xy^1 - yy^1 - x = 0$$

$$yy^1[y^1 - 1] + x[y^1 - 1] = 0$$

$$\frac{dy}{dx} = 1 \quad \text{or} \quad \frac{dy}{dx} = -\frac{x}{y}$$

$$y = x + c \quad \text{or} \quad y^2 = -x^2 + c$$

as curves passing through (3, 4)

$$y = x + 1$$

$$\text{and } x^2 + y^2 = 25$$

12. Ans. (B,C,D)

$$f'(1^+) = \lim_{h \rightarrow 0} \frac{e^{\tan^{-1} h} + h^{2017} \cdot \sin^2 \frac{1}{h} - 1}{h}$$

$$= \lim_{h \rightarrow 0} \left(\left(\frac{e^{\tan^{-1} h} - 1}{\tan^{-1} h} \cdot \frac{\tan^{-1} h}{h} \right) + \left(h^{2016} \cdot \sin^2 \frac{1}{h} \right) \right)$$

$$= 1 + 0 = 1$$

similarly $f'(1^-) = 1 \Rightarrow f'(1) = 1$

Now, $\ell = \lim_{n \rightarrow \infty} \left(n \sum_{k=1}^{10} f\left(1 + \frac{k}{n}\right) - 10n \right)$

$$= \lim_{n \rightarrow \infty} n \left(\sum_{k=1}^{10} f\left(1 + \frac{k}{n}\right) - 10 \right)$$

$$= \lim_{n \rightarrow \infty} \frac{f\left(1 + \frac{1}{n}\right) - f(1)}{\frac{1}{n}} + 2 \left(\frac{f\left(1 + \frac{2}{n}\right) - f(1)}{\frac{2}{n}} \right) + \dots + 10 \left(\frac{f\left(1 + \frac{10}{n}\right) - f(1)}{\frac{10}{n}} \right)$$

$$= f'(1) + 2f'(1) + \dots + 10f'(1) = 55f'(1) = 55$$

SECTION-II

1. Ans. 0.00

$\bar{x}$, $\bar{y}$ & $\bar{x} \times \bar{y}$ are linearly independent

$$a\alpha^2 + b\alpha + c = 0$$

$$a\beta^2 + b\beta + c = 0$$

$$a\gamma^2 + b\gamma + c = 0$$

Quadratic becomes $ax^2 + bx + c = 0$

which is an identity and $a + b + c = 0$

2. Ans. 7.00

Sol. Let $P(x) = ax^2 + bx^2 + cx^2 + dx + e$

$$\therefore \lim_{x \rightarrow 0} \left(\frac{P(x)}{x} - 12 \right) = 0$$

$$\Rightarrow d = 12 \text{ \& } e = 0$$

$$P(x) = ax^4 + bx^3 + cx^2 + 12x$$

$$P(1) = a + b + c + 12 = 25$$

$$\Rightarrow a + b + c = 13 \quad \dots\dots(i)$$

$$P(-1) = a - b + c - 12 = -7$$

$$\Rightarrow a - b + c = 5 \quad \dots\dots(ii)$$

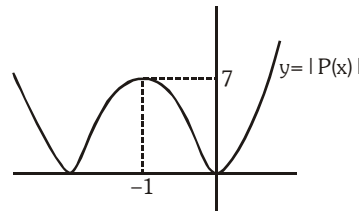
$$P'(x) = 4x^3 + 3bx^2 + 2cx + 12$$

$$\Rightarrow -4a + 3b - 2c = -12 \quad \dots\dots(iii)$$

Solving (i), (ii) & (iii)

$$a = 3, b = 4 \text{ \& } c = 6$$

$$\therefore P(x) = 3x^4 + 4x^3 + 6x^2 + 12x$$



$$\Rightarrow k = 7$$

3. Ans. 1.12

Sol. $I_2 = (\cos x)^{\sqrt{17}+4} (\sin x)^{\pi/2}$

$$- (\sqrt{17} + 4) \int (\cos x)^{\sqrt{17}+3} (\cos^2 x - 1) dx$$

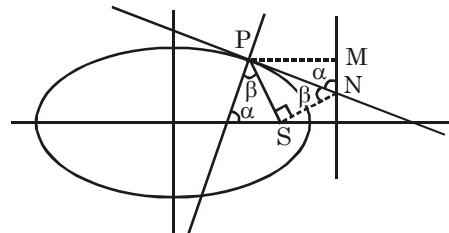
$$\Rightarrow I_2 = 0 - (\sqrt{17} + 4)(I_2 - I_1)$$

$$\Rightarrow I_2 (\sqrt{17} + 5) = (\sqrt{17} + 4) I_1$$

$$\Rightarrow \frac{I_1}{I_2} = 1 + \frac{1}{\sqrt{17} + 4} \Rightarrow \frac{I_1}{I_2} = 1 + \sqrt{17} - 4$$

$$= \sqrt{17} - 3$$

4. Ans. 0.20



$$\sin \alpha = \frac{PM}{PN}$$

$$\sin \beta = \frac{PS}{PN}$$

$$\frac{\sin \beta}{\sin \alpha} = \frac{PS}{PM} = \frac{1}{3}$$

$$\sin \beta = \frac{1}{3} \cdot \sin \alpha = \frac{1}{3} \cdot \frac{3}{5} = \frac{1}{5}$$

5. Ans. 36.00

$$A = (d_1, d_2, d_3, d_4)$$

$$A^4 = (d_1^4, d_2^4, d_3^4, d_4^4) = I$$

$$\Rightarrow d_1^4 = d_2^4 = d_3^4 = d_4^4 = I$$

$$\Rightarrow d_1, d_2, d_3, d_4 \text{ are fourth roots of unity as } d_1 + d_2 + d_3 + d_4 = 0$$

$\Rightarrow \left(2 \cdot \frac{4!}{2!2!}\right) + 4! = 36$ ways are there to assign values to d_1, d_2, d_3, d_4 .

Also $d_1 d_2 d_3 d_4$ is product of 4th roots of unity which is -1 or 1

when $1, -1, 1, -1$ or $i, -i, i, -i$ are used.

6. Ans. 194.00

$$a_1 = 2, \frac{a_n}{a_{n-1}} = \frac{a_{n-1}}{a_{n-2}}$$

$\Rightarrow a_1, a_2, a_3, \dots, a_n$ are in G.P.

Let $a_2 = x$, then for $x = 3$,

$$\frac{a_3}{a_2} = \frac{a_2}{a_1} = \frac{x^2}{2}$$

$$\Rightarrow a_1^2 = a_1 a_3 \Rightarrow a_3 = \frac{x^2}{2}$$

$\Rightarrow 2, x, \frac{x^2}{2}, \frac{x^3}{4}, \frac{x^4}{8}, \dots$ with common ratio,

$$r = \frac{x}{2}$$

$$\text{given } \frac{x^4}{8} \leq 162 \Rightarrow x \leq 6$$

Also x & $\frac{x^4}{8}$ are integer's, so if x is even

then only $\frac{x^4}{8}$ will be an integer. So only possible value of x is 4 and 6.